

PERSONALIZED CHEAT SHEET

1. BASIC LINUX COMMANDS

Command	Sample Usage / Description
pwd	Show the full path of the current working directory
ls -l	List files in the current folder with details (permissions, size, date)
cd Documents	Move into the 'Documents' folder
mkdir project	Create a new folder named 'project'
touch notes.txt	Create a new empty file named 'notes.txt'
cp file.txt backup.txt	Copy 'file.txt' and save it as 'backup.txt'
mv old.txt new.txt	Rename or move 'old.txt' to 'new.txt'
rm -r old-folder	Delete a folder and everything inside it
cat notes.txt	Print the contents of 'notes.txt' to the screen
grep "error" log.txt	Search 'log.txt' for lines containing the word 'error'
chmod +x setup.sh	Make the file 'setup.sh' executable
df -h	Show disk space usage in a human-readable format

2. BASIC GIT COMMANDS

Command	Sample Usage	Description	Common Flags
git init	git init	Turns the current folder into a new, empty Git repository	--bare
git status	git status	Shows which files are staged, unstaged, or untracked	-s
git add	git add index.html	Moves a file's changes into the staging area for the next commit	-A, .
git commit	git commit -m "fix bug"	Saves the staged changes permanently into the project history	-m, --amend
git log	git log --oneline	Displays the history of commits made in the repository	--oneline, -p
git diff	git diff	Shows the exact line-by-line changes between file versions	--staged
git restore	git restore index.html	Reverts a file in the working directory back to its last commit	--staged
git branch	git branch feature-x	Lists existing branches or creates a new one	-a, -d, -m
git checkout / switch	git switch feature-x	Moves you to a different branch or restores files	-b, -c
git merge	git merge feature-x	Combines the changes of one branch into the current branch	--no-ff
git tag	git tag -a v1.0	Marks a specific commit with a readable label, like a version number	-a, -d
git clone	git clone <url>	Downloads a full copy of a remote repository to your computer	--depth 1, -b

GIT'S THREE-TREE ARCHITECTURE

1. WORKING DIRECTORY	2. STAGING AREA (INDEX)	3. REPOSITORY (.git)
The folder where you create, modify, and delete project files.	The area where you select and prepare changes before committing them.	The Git storage area that keeps the project's commit history and saved versions.

Flow: `git add` moves changes from the Working Directory into the Staging Area; `git commit` moves staged changes into the Repository.

REFERENCES

- Git. (n.d.). *Git Reference*. <https://git-scm.com/docs>
- Chacon, S., & Straub, B. (n.d.). *Pro Git: Recording Changes to the Repository*. <https://git-scm.com/book/en/v2/Git-Basics-Recording-Changes-to-the-Repository>
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